

Homework 2

Packages

Question 1

```
install.packages("aRtsy")
```

Explanation

The difference between `install.packages` and `library` is that `install.packages` installs whatever package you want to use to your computer, but `library` allows you to use it in whatever file you are working in.

Question 2

```
library("aRtsy")
```

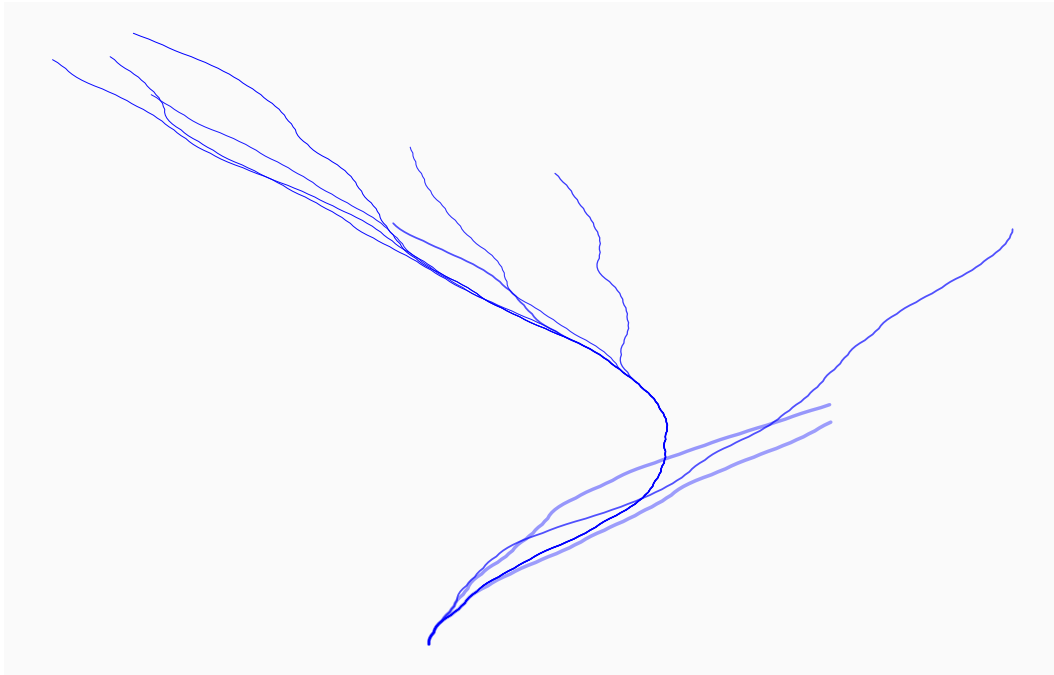
Collatz conjecture

The Collatz conjecture is a famous unsolved math problem that asks if repeating two simple arithmetic steps will always reach the number 1.

$$\text{even} = 3n + 1, \text{odd} = n/2$$

Question 3

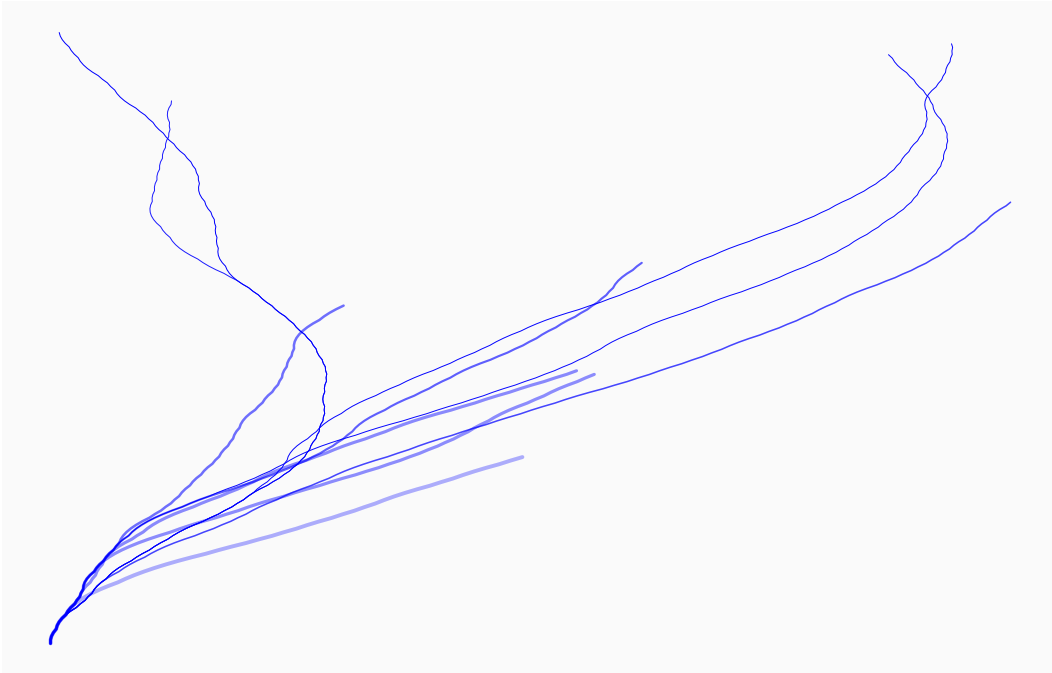
```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



This function provides a customizable canvas. You are allowed to change the color and number of patterns provided. N is the starting number.

part B

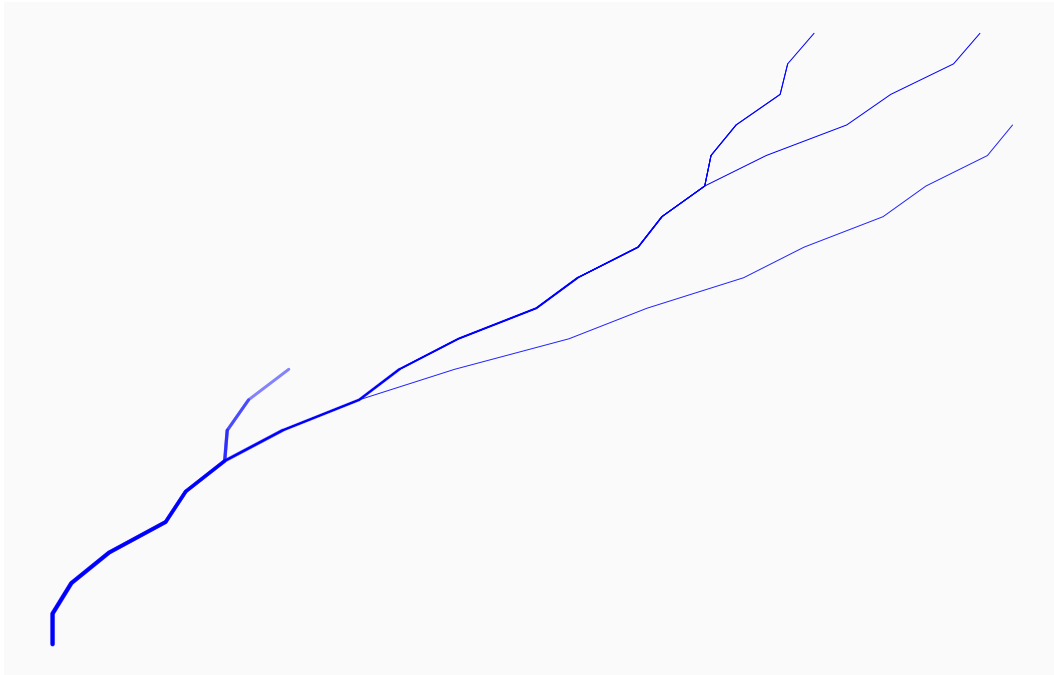
```
set.seed(3)  
canvas_collatz(colors = c("blue"),  
  n = 10  
)
```



- The function displays a different pattern everytime, but every line has the same starting point. This is because $N = 10$, so the function is going to create 10 random patterns all from the same starting point.
- This could be troublesome due to the fact that it does not have the same output every time, meaning if you want something specific to happen then that wouldn't be possible with this type of code.
- `set.seed` can be used to get the same output by producing the same “random” output every time you run the code.

Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



- The difference in the code chunks are that your setting n to a specific number (10) in the first one and you are setting n to a range in the second (1:20)
- No, it seems like the output is reproducing the same every time, so that would defeat the purpose of needing set.seed.